

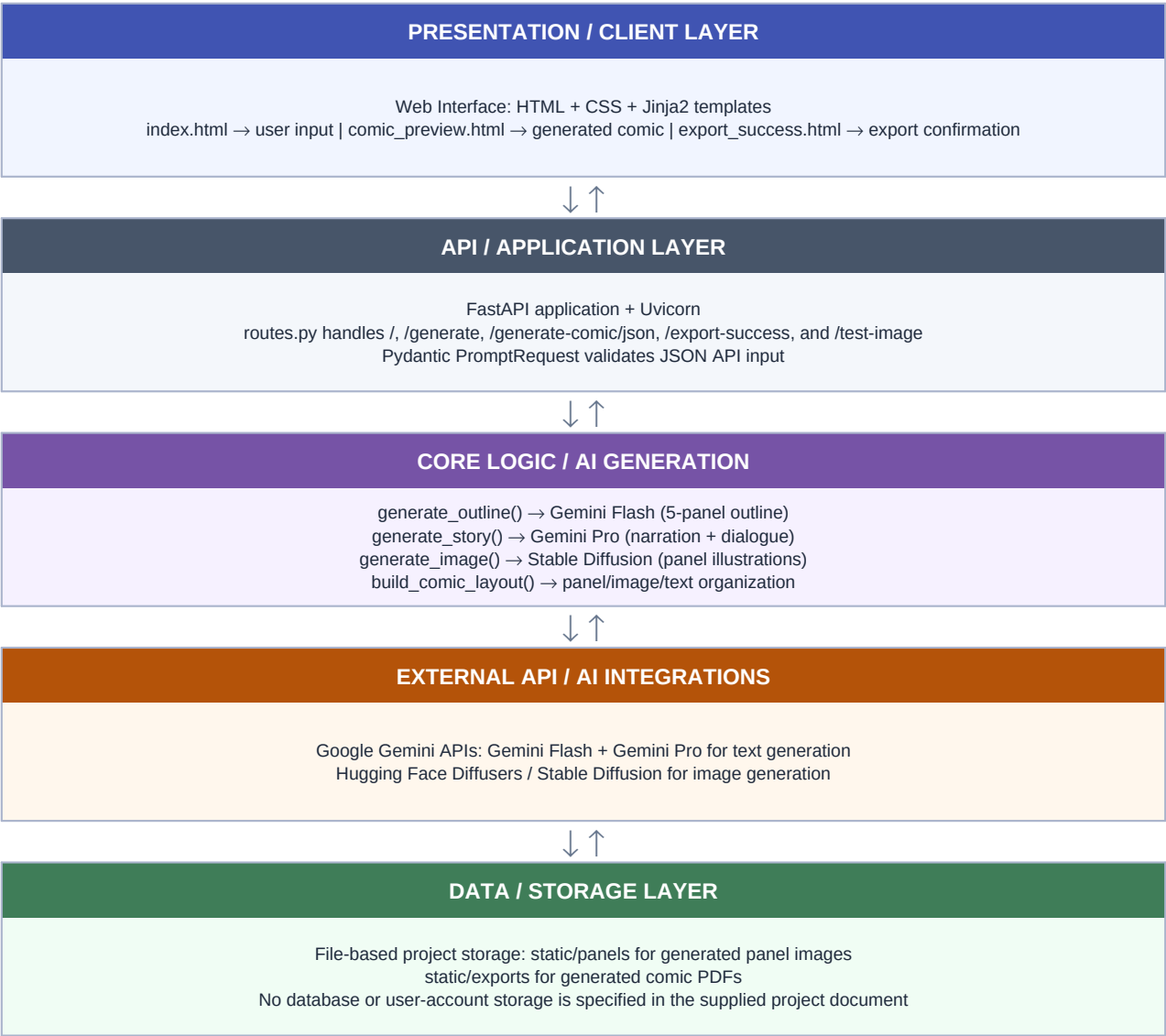
SOLUTION ARCHITECTURE

ComicCraft – AI Comic Story Creator using Gemini Models

Date	15 March 2026	Team ID	Not provided
Project Name	ComicCraft – AI Comic Story Creator using Gemini Models	Maximum Marks	5 Marks

Solution Architecture Diagram

High-level architecture mapped to the supplied Presentation Layer, Application Layer, External Integration, and Data/Storage structure.



Architecture flow: User enters story details → FastAPI receives and validates input → Gemini models generate outline/story → Stable Diffusion generates images → layout builder combines content → FPDF exports the complete comic as PDF → frontend shows the result/download confirmation.

Component Description Table

The following table fills the component template using only technologies and responsibilities described in the supplied ComicCraft project documentation.

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects story prompt, character name, setting, tone and art style; displays generated comic panels and export status.	HTML, CSS, Jinja2
API / Application Layer	Receives form or JSON input, coordinates AI generation, handles routing, error handling, template rendering and PDF export.	FastAPI, Uvicorn, Pydantic, routes.py
Core Logic Service	Creates the structured 5-panel outline, expands it into narration/dialogue, generates images, and builds the final comic layout.	Gemini Flash, Gemini Pro, Stable Diffusion, Python modules
External API Integrations	Provides generative AI capabilities for structured story content, detailed creative writing, dialogue and comic-style illustrations.	Google Gemini APIs; Hugging Face Diffusers / Stable Diffusion
Data / Storage Layer	Stores generated panel images and exported PDF files for preview/download. The supplied project does not specify a database.	static/panels, static/exports, FPDF
Authentication Service	Not specified in the current project architecture. The project documentation states that the current version focuses on single comic generation without user accounts.	Not specified

Technology & Workflow Summary

Stage	Implementation
Input	Story prompt, character name, setting, tone, art style
Story outline	Gemini Flash generates a structured 5-panel outline
Narration	Gemini Pro generates narration and character dialogue
Illustration	Stable Diffusion generates comic-style images for each panel
Layout	build_comic_layout() pairs images with panel text
Export	save_pdf() uses FPDF to create the downloadable multi-page PDF
Frontend flow	index.html → comic_preview.html → export_success.html

Source basis: Project details are derived from the supplied ComicCraft project document; the structure follows the supplied Solution Architecture template. No additional technologies, database, authentication system, or deployment platform have been assumed where they were not specified.